

AMAL A K

Full-Stack Software Engineer | React · Node.js · TypeScript · AI Applications · SaaS
Trivandrum, India | +91 9605655082 | amal.anandvijay@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with production experience in React.js, Next.js, Node.js, Express.js, PostgreSQL, MongoDB, and TypeScript. Built a multi-tenant healthcare SaaS with database-per-tenant isolation, automated cloud provisioning, and Amazon Bedrock RAG integration. Developed AI-powered applications using Google Gemini and Tesseract.js OCR. Contributed to production applications across healthtech, e-commerce, and edtech domains.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript

Frontend: React.js, Next.js, Redux, Redux Toolkit, Tailwind CSS, Shadcn/UI, Material UI, Recharts, Vite

Backend: Node.js, Express.js, RESTful APIs, JWT Authentication, Multer

Databases: PostgreSQL, MongoDB Atlas, Prisma ORM, Sequelize, Mongoose, NeonDB

Cloud & DevOps: AWS S3, Cloudflare DNS, Vercel, Render, Netlify, Docker (basic), CI/CD

AI & Integrations: Amazon Bedrock (LLM, RAG), Google Gemini API (LLM), Tesseract.js (OCR), pdf-parse

Tools: Git, GitHub, Postman, Axios, VS Code

Architecture: Multi-Tenant SaaS, Microservices, Layered MVC, RBAC, System Design, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer Trainee | Softnotions Technologies, Technopark, Trivandrum

Jan 2026 – Apr 2026

- Built **DiagnomiQ**, a production multi-tenant healthcare SaaS (PERN stack) with Database-per-Tenant architecture — automated PostgreSQL provisioning, Cloudflare DNS subdomain setup, and AWS S3 folder-level isolation per tenant.
- Implemented JWT-based auth with RBAC and custom Tenant DB Middleware for secure, dynamic request routing across isolated per-tenant databases.
- Integrated Amazon Bedrock RAG pipeline with a structured medical knowledge base to deliver real-time AI-powered clinical decision support in the live platform.

MERN Stack Developer | Relaxplzz Technologies, Technopark, Trivandrum

Apr 2025 – Aug 2025

- Developed frontend features and RESTful APIs for **Giftomart**, a production e-commerce platform, using React.js, TypeScript, Tailwind CSS, MongoDB, Node.js, and Express.js.
- Built a reusable TypeScript component library, improving maintainability; collaborated with product, design, and QA teams across Agile sprints.

MERN Stack Developer | SciPy Technologies, Trivandrum

Jun 2023 – Jan 2024

- Developed Node.js RESTful APIs and React UIs with Redux for a microservices-based ERP system serving an educational institution.
- Designed MongoDB schemas with Mongoose, integrated third-party APIs via Axios, and validated endpoints with Postman.

FEATURED PROJECTS

DiagnomiQ — Multi-Tenant Healthcare SaaS | Company Project

React · TypeScript · Node.js · PostgreSQL · Sequelize · AWS S3 · Cloudflare · JWT · Amazon Bedrock

- Database-per-Tenant architecture: automated PostgreSQL provisioning, Cloudflare subdomain creation, and AWS S3 folder isolation for HIPAA-aligned data separation per tenant.
- Amazon Bedrock RAG pipeline with structured medical knowledge base for AI-powered clinical decision support.
- Oncology risk dashboards (Recharts) with IPI and aalPI survival probability charts for real-time lymphoma patient risk stratification.

TripMind AI — AI Travel Planner | [GitHub](#) · [Live Demo](#)

React · TypeScript · Node.js · Express.js · MongoDB Atlas · Google Gemini API · Tesseract.js · JWT · Vercel · Render

- OCR and PDF extraction pipeline (Tesseract.js + pdf-parse) feeds booking data into Google Gemini to generate structured, day-by-day travel itineraries.
- Layered Node.js backend (Routes → Controllers → Services → Models) with JWT auth, MongoDB Atlas, and Gemini retry logic — exponential backoff with automatic model fallback.
- Public itinerary sharing via unique share links, expiring password-reset tokens; deployed full-stack on Vercel and Render.

FinSight — Personal Finance Tracker | [GitHub](#) · [Live Demo](#)

Next.js · TypeScript · PostgreSQL · Prisma ORM · Tailwind CSS · Shadcn/UI · Vercel

- Type-safe full-stack finance tracker (Next.js + Prisma ORM) with interactive analytics dashboards, category-based filtering, and Vercel edge deployment.
- End-to-end TypeScript with Prisma-generated types across frontend and backend, improving reliability and developer experience.

EDUCATION

Master of Computer Applications (MCA) | Indira Gandhi National Open University

2021 – Present

B.Sc. Computer Science | Emmanuel College, Vazhichal

2018 – 2021